

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10	U04						*****VALVES*****			812	NIPPLE TO BE CUT FROM SS SCH XXS PIPE.
20										816	COMPACT FLANGES CAN BE USED FOR TIE-IN PURPOSE ONLY TO EXISTING PIPING AND EQUIPMENT.
30	U04						***** MODULAR VALVES *****				
30	U04		MODULAR DBB	K15AS22	0.5	1.5	MODULAR VALVE DB&B BALL TYPE 1500# RF SS A182 F316 316 SS TRIM METAL LO 718 BOLT - FULL SIZE TYPE 2BV	0	0	817	USE ONLY FOR TIE IN
40	U04		MODULAR DBB	K15AS26	2	2	MODULAR VALVE DB&B BALL TYPE 1500# RF SS A182 F316 316 SS TRIM METAL LO 718 BOLT - FULL SIZE TYPE 2BV	0	0	B1	ALL BOLTS ARE METRIC DIA.
50	U04	VAV, 817	MODULAR DBB	K15CS26	3	3	MODULAR VALVE DB&B BALL TYPE 1500# CF SS A182 F316 316 SS TRIM METAL LO 718 BOLT - FULL SIZE TYPE 2BV	0	0	B19	1" X 160MM BOLTS ONLY TO BE USED FOR 1-1/2 1500# PERAR BALL VALVES ITEM CODE B15AS21.
60	U04		MODULAR DBB	M15AS22	0.75	1.5	MODULAR VALVE DB&B BALL TYPE 1500# RF SS A182 F316 316 SS TRIM METAL LO 718 BOLT - CONTROLLED BORE TYPE 2BVR	0	0	B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10
70	U04		MODULAR DBB	M15AS26	2	2	MODULAR VALVE DB&B BALL TYPE 1500# RF SS A182 F316 316 SS TRIM METAL LO 718 BOLT - CONTROLLED BORE TYPE 2BVR	0	0	B20	1" X 175MM BOLTS ONLY TO BE USED FOR 1-1/2 1500# PERAR BALL VALVES ITEM CODE B15AS21 WHEREVER MATCHED WITH A SPECTACLE BLIND
80	U04	VAW	MODULAR DBB	M15ES22	0.75	1	MODULAR VALVE DB&B BALL TYPE 1500# RF SS A182 F316SS TRIM METAL LO 718 BOLT REDUCING - CONTROLLED BORE TYPE 2BVR	2	2		
90	U04	VAW, 817	MODULAR DBB	M15ZS22	1	1	MODULAR VALVE DB&B BALL TYPE 1500# CF x RF SS A182F316 316SS TRIM METAL LO REDUCING - CONTROLLED BORE 718 BOLT TYPE 2BVR	3	3	B22	L7M BOLTS SHALL ONLY BE USED IN SOUR SERVICE LINES WITH DESIGN TEMPS GREATER THAN OR EQUAL TO -46 DEG C AND AT LOCATIONS WHERE FLANGE JOINTS ARE INSULATED
100							***** BALL VALVES *****				
100	U04										
110											
110	U04		BALL	A15AS27	2	2	BALL 1500# RF SS A182 F316 316SS TRIM TRUNNION SOFT RB GO 718 BOLT	0	0	B23	B8M CL.2 BOLTS SHALL BE USED BELOW -46 DEG C. ENGINEERING SHALL BE CONSULTED.
120	U04	817	BALL	A15CS27	4	4	BALL 1500# CF SS A182 F316 316SS TRIM TRUNNION SOFT RB GO 718 BOLT	0	0		
130	U04		BALL	B15AS21	0.75	1.5	BALL 1500# RF SS A182 F316 316SS TRIM FLOATING SOFT FB LO 718 BOLT	0	0		PRIOR TO USING B8M CL.2 BOLTS IN SOUR SERVICE LINES AT LOCATIONS WHERE FLANGE JOINTS ARE INSULATED.
140	U04	VA5	BALL	B15AS27	2	2	BALL 1500# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO 718 BOLT	0	0		
150	U04	817	BALL	B15CS27	4	4	BALL 1500# CF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO 718 BOLT	0	0		
160							***** GLOBE VALVES *****			BA4	TO BE USED WITH COMPACT FLANGES REQUIRING STANDARD NUTS ONLY.
160	U04									FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.
170	U04		GLOBE	L15AS55	0.75	1.5	GLOBE 1500# RF A182 F316 SS TRIM 16 OS&Y HW 718 BOLT	0	0		
180	U04		GLOBE	L15AS56	2	2	GLOBE 1500# RF SS A351 CF8M TRIM 16 OS&Y HW 718 BOLT	0	0	UJ2	USE 1 1/2IN WITH THERMOWELL
190	U04	817	GLOBE	L15CS56	4	4	GLOBE 1500# CF SS A351 CF8M TRIM 16 OS&Y GO 718 BOLT	0	0		CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.
200							***** CHECK VALVES *****				
200	U04									VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
210											
210	U04	VC1	CHECK	C15AS54	0.75	1.5	CHECK PISTON 1500# RF SS A182 F316 TRIM 16 718 BOLT	0	0		USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.
220	U04		CHECK	C15AS45	2	2	CHECK DUAL WAFER LUG 1500# RF SS A351 CF8M TRIM 16 THROU BOLTED	0	0	VAV	
230	U04	817	CHECK	C15CS52	4	4	CHECK DUAL DBL FLG 1500# CF SS A351 CF8M TRIM 16	0	0		VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.
240							***** PIPE *****				
240	U04										
250											
250	U04		PIPE	5365207	0.75	1.5	PIPE SCH 160 SMLS SS A312 TP316/316L PE	0	0	VC1	SPRING PISTON CHECK VALVES ARE PREFERRED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.
260	U04		PIPE	5365205	2	2	PIPE SCH 160 SMLS SS A312 TP316/316L BE	0	0		
270	U04		PIPE	5365206	4	4	PIPE SCH XXS SMLS SS A312 316/316L SS BE	0	0		
280	U04	812	PIPE	5365206	0.75	0.75	PIPE SCH XXS SMLS SS A312 316/316L SS BE	0	0		
290							***** FITTINGS *****				
290	U04										
300											
300	U04		ELBOW	5449140	0.75	2	ELL BW 90 DEG LR SCH 160 SS A403 WP316/316L-S	0	0		
310	U04		ELBOW	5449142	0.75	2	ELL BW 45 DEG LR SCH 160 SS A403 WP316/316L-S	0	0		
320											
320	U04		TEE	5449190	0.75	2	TEE SCH 160 SS A403 WP316/316L-S	0	0		
330	U04		TEE RED	5449686	0.75	1.5	TEE RED SCH 160 316/316L SS A403	1	2		
340											
340	U04		CAP	5449055	0.75	2	CAP SCH 160 SS A403 WP316/316L-S	0	0		
350											
350	U04		RED CONC	5448912	1.5	2	REDUCER CONC SCH XXS/160 SS A403 WP 316/316L-S	4	4		
360	U04		RED CONC	5449689	0.75	1	REDUCER CONC SCH 160 SS A403 WP 316/316L-S	1	2		
370	U04		RED ECC	5449187	1.5	2	REDUCER ECC SCH XXS/160 SS A403 WP316/316L-S	4	4		
380	U04		RED ECC	5449690	0.75	1	REDUCER ECC SCH 160 SS A 403 WP 316/316L-S	1	2		
390											
390	U04	UJ2	OLETS	5504048	1.5	2	NIPOFLANGE 1500# RF SCH XXS/160 SS A182 F316/316L	4	4		
400											
400	U04	UJ2	ELBOFLG	5516439	1.5	1.5	ELBOFLANGE 1500# RF SCH XXS/160 A182 F316/316L	4	4		
410							***** FLANGES *****				
410	U04	816									
420											
420	U04	FA8	FLANGE	5600664	0.5	0.5	FLG SCRDR (FNPT) REDUCING 1500# RF SS A182 F316/316L	0.75	1.5		
430											
430	U04		FLG ADAPTER	5851205	0.75	0.75	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE	0	0		
440	U04		FLG ADAPTER	5851205	2	2	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE	0	0		
450											
450	U04		FLANGE	5583003	0.75	2	FLG WN 1500# RF SS A182 F316/316L SCH 160 BORE	0	0		

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES BUTT - WELDED SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - NONE	CORR. ALLOW. 1 MM	SOUR SERVICE NACE MR0175/ISO 15156 DESIGN: ASME B16.5 MG 2.2 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: 316/316L SS DUAL GRADE FLANGE RATING: 1500 RF/CF TEMPERATURE RANGE: -100 / 200°C	12/16/2024	BY OZAC	U04	Page 1 of 2	1500P2

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460	U04	817	FLANGE	5619840	4	4	FLG COMPACT 1500# RJ SS A182 F316/316L SCH XXS LINE CLASS 1500P1	0	0	
470	U04		BLIND	5616564	0.5	2	FLG BLIND 1500# RF SS A182 F316/316L	0	0	
480	U04	817	BLIND	5619789	4	4	FLG COMPACT BLIND 1500# RJ SS A182 F316/316L LINE CLASS 1500P1	0	0	
490										
490	U04						***** ORIFICE FLANGES *****			
500										
500	U04		ORIFICE FLG	5641520	2	2	WN ORIF SET 1500# RF SS A182 F316/316L SCH 160 BORE	0	0	
510										
510	U04						***** PIPE NIPPLES *****			
520										
520	U04	812					NIPPLE BBE CUT FROM PIPE			
530										
530	U04						***** GASKETS *****			
540										
540	U04		GASKET	5672251	0.5	2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
550	U04	817	GASKET	5673490	4	4	GASKET 1500# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE LINE CLASS 1500P1	0	0	
560										
560	U04	B1					***** BOLTING *****			
570										
570	U04	B23	BOLTING	5675677	0.5	2	STUD-BOLT A320 GR B8M CLASS 2/A194 GR 8MA NUTS	0	0	
580	U04	BA4	BOLTING	5675776	3	4	STUD-BOLT WITH 2 NUTS API-6A718	0	0	
590										
590	U04	B19		5675663	1.5	1.5	IMPERIAL STUD-BOLT A320 GR B8M CLASS 2/A194 GR 8MA 2 NUTS 1" X 160MM X SET OF 4 OFF	0	0	
600	U04	B20		5675664	1.5	1.5	IMPERIAL STUD-BOLT A320 GR B8M CLASS 2/A194 GR 8MA 2 NUTS 1" X 175MM X SET OF 4 OFF	0	0	
610										
610	U04	B22		5675770	0.5	2	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
620										
620	U04						***** BLINDS & SPACERS *****			
630										
630	U04		SPEC BLIND	5749757	1	2	SPECTACLE BLIND 1500# RF SS A240 GR 316	0	0	
640										

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